```
Last login: Sun Jun 28 07:08:14 on ttys000
stephenrajva@Stephens-MacBook-Air-4 ~ % cd "/Users/stephenrajva/Downloads/HPC-Up
load 2/SST_Stimulator"
echo "===== folders matching Communication ====="
ls -d *ommunication* 2>/dev/null
echo ""
echo "===== Communication_Pattern_PPN1 (the PPN=1 study) ====="
ls -la "Communication_Pattern_PPN1" 2>/dev/null || echo "not found under that ex
act name"
echo ""
echo "===== the PPN=1 pattern config + analyzer ====="
cat "Communication_Pattern_PPN1/pattern_mesh_ppn1.py" 2>/dev/null | head -80 ||
find . -name "pattern_mesh_ppn1.py" 2>/dev/null
===== folders matching Communication =====
Communication _Pattern

===== Communication_Pattern_PPN1 (the PPN=1 study) =====
not found under that exact name

===== the PPN=1 pattern config + analyzer =====
stephenrajva@Stephens-MacBook-Air-4 SST_Stimulator % cd "/Users/stephenrajva/Dow
nloads/HPC-Upload 2"
echo "===== searching for pattern configs anywhere ====="
find . -name "pattern_mesh*.py" 2>/dev/null
find . -name "*pattern*ppn1*.py" 2>/dev/null
find . -name "analyze_patterns.py" 2>/dev/null
echo ""
echo "===== what's actually inside the spaced folder ====="
ls -la "SST_Stimulator/Communication _Pattern"
===== searching for pattern configs anywhere =====
./SST_Stimulator/Communication _Pattern/Communication_Pattern_PPN1/pattern_mesh_
ppn1.py
./SST_Stimulator/Communication _Pattern/Communication_Pattern_PPN1/pattern_mesh_
ppn1.py
./SST_Stimulator/Communication _Pattern/Communication_Pattern_PPN1/analyze_patte
rns.py

===== what's actually inside the spaced folder =====
total 168
drwxr-xr-x  11 stephenrajva  staff    352 28 Jun 07:17 .
drwxr-xr-x@  7 stephenrajva  staff    224 28 Jun 07:17 ..
-rw-r--r--@  1 stephenrajva  staff  14340 28 Jun 07:17 .DS_Store
drwxr-xr-x  15 stephenrajva  staff    480 17 Jun 11:54 adjacent_experiment
-rw-r--r--   1 stephenrajva  staff    125 18 Jun 11:55 communication_comparison.
csv
-rw-r--r--   1 stephenrajva  staff  38889 18 Jun 11:56 communication_comparison.
png
drwxr-xr-x  12 stephenrajva  staff    384 25 Jun 01:46 Communication_Pattern_PPN
1
-rw-r--r--@  1 stephenrajva  staff  20268 18 Jun 11:59 Comparison_terminal.pdf
drwxr-xr-x   4 stephenrajva  staff    128 18 Jun 11:57 HalfShift_experiment
drwxr-xr-x   4 stephenrajva  staff    128 18 Jun 11:57 Mirrored_experiment
-rw-r--r--   1 stephenrajva  staff    575 18 Jun 11:56 plot_comparison.py
stephenrajva@Stephens-MacBook-Air-4 HPC-Upload 2 % cd "/Users/stephenrajva/Downl
oads/HPC-Upload 2/SST_Stimulator/Communication _Pattern/Communication_Pattern_PP
N1"
echo "===== folder contents ====="
ls -la
```

```
echo ""
echo "===== pattern_mesh_ppn1.py ====="
cat pattern_mesh_ppn1.py
echo ""
echo "===== run script(s) ====="
cat run_pattern_sweep.sh 2>/dev/null || ls *.sh 2>/dev/null
===== folder contents =====
total 488
drwxr-xr-x  12 stephenrajva  staff    384 25 Jun 01:46 .
drwxr-xr-x  11 stephenrajva  staff    352 28 Jun 07:17 ..
-rw-r--r--@  1 stephenrajva  staff  10244 25 Jun 01:15 .DS_Store
drwxr-xr-x@  4 stephenrajva  staff    128 24 Jun 23:41 adjacent
-rw-r--r--@  1 stephenrajva  staff   3518 25 Jun 00:03 analyze_patterns.py
drwxr-xr-x   4 stephenrajva  staff    128 25 Jun 00:49 halfshift
drwxr-xr-x   3 stephenrajva  staff     96 24 Jun 23:57 mirrored
-rw-r--r--   1 stephenrajva  staff  66282 25 Jun 00:04 pattern_hops_vs_size.png
-rw-r--r--   1 stephenrajva  staff  87833 25 Jun 00:04 pattern_latency_vs_size.p
ng
-rw-r--r--@  1 stephenrajva  staff   3580 24 Jun 23:56 pattern_mesh_ppn1.py
-rw-r--r--@  1 stephenrajva  staff    568 24 Jun 23:39 run_pattern_sweep.sh
-rw-r--r--@  1 stephenrajva  staff  61799 25 Jun 01:46 Untitled.pdf

===== pattern_mesh_ppn1.py =====
#!/usr/bin/env python
# ============================================================================
#  pattern_mesh_ppn1.py  -  Communication-pattern distance signature on a Mesh
#  PPN = 1.  Single job, linear allocation (rank i -> node i), one PingPong
#  between rank 0 and its pattern-partner. Measures the characteristic-distance
#  latency/hops for each pattern as the mesh grows.
#
#    adjacent  -> rank 0 <-> rank 1      (nearest neighbour)
#    halfshift -> rank 0 <-> rank N/2    (half-way across)
#    mirrored  -> rank 0 <-> rank N-1    (opposite corner)
#
#  USAGE: export MESH_SHAPE="4x4"; export PATTERN="halfshift"; sst pattern_mesh_
ppn1.py
# ============================================================================

import sst, os
from sst.merlin.base import *
from sst.merlin.endpoint import *
from sst.merlin.interface import *
from sst.merlin.topology import *
from sst.ember import *

shape    = os.environ.get("MESH_SHAPE", "4x4")
pattern  = os.environ.get("PATTERN", "adjacent").lower()
out_dir  = os.environ.get("STATS_DIR", ".")
msg_size = 256
iters    = 35

dims = [int(x) for x in shape.lower().split("x")]
N = 1
for d in dims:
    N *= d                       # nodes == ranks (PPN=1)

if   pattern == "adjacent":  partner = 1
elif pattern == "halfshift": partner = N // 2
```

```
elif pattern == "mirrored":  partner = N - 1
else: raise SystemExit(f"Unknown PATTERN '{pattern}'")

# representative mesh distance (Manhattan) between node 0 and node `partner`
def coord(nid):
    c, rem = [], nid
    for d in dims:
        c.append(rem % d); rem //= d
    return c
def manhattan(a, b):
    return sum(abs(x - y) for x, y in zip(coord(a), coord(b)))
rep_hops = manhattan(0, partner)

PlatformDefinition.setCurrentPlatform("firefly-defaults")

topo              = topoMesh()
topo.shape        = shape
topo.local_ports  = 1
topo.width        = "x".join(["1"] * len(dims))
topo.link_latency = "25ns"

router                 = hr_router()
router.link_bw         = "12GB/s"
router.flit_size       = "16B"
router.xbar_bw         = "20GB/s"
router.input_latency   = "30ns"
router.output_latency  = "30ns"
router.input_buf_size  = "32kB"
router.output_buf_size = "32kB"
router.num_vns         = 2
router.xbar_arb        = "merlin.xbar_arb_lru"
topo.router = router

nic                 = ReorderLinkControl()
nic.link_bw         = "12GB/s"
nic.input_buf_size  = "16kB"
nic.output_buf_size = "16kB"

system = System()
system.setTopology(topo)

ep = EmberMPIJob(0, N)
ep.network_interface = nic
ep.addMotif("Init")
ep.addMotif(f"PingPong rank2={partner} messageSize={msg_size} iterations={iters}
 compute=3ns")
ep.addMotif("Fini")
ep.nic.nic2host_lat = "100ns"
system.allocateNodes(ep, "linear")
system.build()

csv_path = os.path.join(out_dir, f"{pattern}_mesh_{shape}_ppn1_nodes{N}_stats.cs
v")
sst.setStatisticLoadLevel(10)
sst.setStatisticOutput("sst.statOutputCSV", {"filepath": csv_path, "separator":
", "})
sst.enableAllStatisticsForComponentType("merlin.hr_router")
sst.enableAllStatisticsForComponentType("merlin.linkcontrol")
```

```
sst.enableAllStatisticsForComponentType("ember.EmberEngine", {"MotifLog": "1"})
sst.enableAllStatisticsForComponentType("ember.nic")

print("==================================================")
print(f" Pattern            : {pattern}")
print(f" Topology           : Mesh {shape}  ({N} nodes, PPN=1)")
print(f" Exchange           : rank 0  <->  rank {partner}")
print(f" Representative hops: {rep_hops}  (Manhattan dist on mesh)")
print(f" Stats CSV          : {csv_path}")
print("==================================================")

===== run script(s) =====
#!/usr/bin/env bash
set -e
SCRIPT="$(pwd)/pattern_mesh_ppn1.py"
PATTERNS=("adjacent" "halfshift" "mirrored")
SHAPES=("2x2" "4x4" "8x8" "16x16" "32x32")   # 4,16,64,256,1024 nodes
for P in "${PATTERNS[@]}"; do
    OUT_DIR="$(pwd)/$P/results"
    mkdir -p "$OUT_DIR"
    for SHAPE in "${SHAPES[@]}"; do
        echo "=== $P on mesh $SHAPE (PPN=1) ==="
        export PATTERN="$P"
        export MESH_SHAPE="$SHAPE"
        export STATS_DIR="$OUT_DIR"
        sst "$SCRIPT" 2>&1 | tee "$OUT_DIR/${P}_mesh_${SHAPE}_run.log"
    done
done
echo "All pattern runs finished."
stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 % cd "/Users/step
henrajva/Downloads/HPC-Upload 2/SST_Stimulator/Communication _Pattern/Communicat
ion_Pattern_PPN1"
mkdir -p PPN_2
mv ~/Downloads/pattern_mesh_ppn2.py ~/Downloads/run_pattern_sweep_ppn2.sh PPN_2/
 2>/dev/null
ls -l PPN_2/
total 16
-rw-r--r--@ 1 stephenrajva  staff  4075 28 Jun 07:21 pattern_mesh_ppn2.py
-rw-r--r--@ 1 stephenrajva  staff   598 28 Jun 07:21 run_pattern_sweep_ppn2.sh
stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 % cd "/Users/step
henrajva/Downloads/HPC-Upload 2/SST_Stimulator/Communication _Pattern/Communicat
ion_Pattern_PPN1/PPN_2"
chmod +x run_pattern_sweep_ppn2.sh
export MESH_SHAPE="2x2"; export PATTERN="mirrored"; export STATS_DIR="./adjacent
/results"
sst pattern_mesh_ppn2.py
==================================================
 Pattern            : mirrored
 Topology           : Mesh 2x2  (4 routers, 8 nodes, PPN=2)
 Exchange           : rank 0  <->  rank 7
 Representative hops: 2  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV          : ./adjacent/results/mirrored_mesh_2x2_ppn2_nodes8_stats.csv
==================================================
180000:0:7:EmberEngine:PingPong:generate:65: rank=7 size=8
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=7 messageSize=256 iterations=35
PingPong: otherRank 7, total time 75.319 us, loop 35, bufLen 256, latency 1.076
```

```
us. bandwidth 0.237921 GB/s
Simulation is complete, simulated time: 80.0779 us
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP
C-Upload 2/SST_Stimulator/Communication _Pattern/Communication_Pattern_PPN1/PPN_
2"
./run_pattern_sweep_ppn2.sh
ls -l adjacent/results/*.csv halfshift/results/*.csv mirrored/results/*.csv
=== adjacent on mesh 2x2 (PPN=2) ===
==================================================
 Pattern           : adjacent
 Topology          : Mesh 2x2  (4 routers, 8 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 1
 Representative hops: 0  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/adjacent/results/adjacent
_mesh_2x2_ppn2_nodes8_stats.csv
==================================================
180000:0:1:EmberEngine:PingPong:generate:65: rank=1 size=8
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=1 messageSize=256 iterations=35
PingPong: otherRank 1, total time 63.223 us, loop 35, bufLen 256, latency 0.903
us. bandwidth 0.283441 GB/s
Simulation is complete, simulated time: 66.9459 us
=== adjacent on mesh 4x4 (PPN=2) ===
==================================================
 Pattern           : adjacent
 Topology          : Mesh 4x4  (16 routers, 32 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 1
 Representative hops: 0  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/adjacent/results/adjacent
_mesh_4x4_ppn2_nodes32_stats.csv
==================================================
180000:0:1:EmberEngine:PingPong:generate:65: rank=1 size=32
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=1 messageSize=256 iterations=35
PingPong: otherRank 1, total time 63.223 us, loop 35, bufLen 256, latency 0.903
us. bandwidth 0.283441 GB/s
Simulation is complete, simulated time: 68.9779 us
=== adjacent on mesh 8x8 (PPN=2) ===
==================================================
 Pattern           : adjacent
 Topology          : Mesh 8x8  (64 routers, 128 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 1
 Representative hops: 0  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/adjacent/results/adjacent
_mesh_8x8_ppn2_nodes128_stats.csv
==================================================
180000:0:1:EmberEngine:PingPong:generate:65: rank=1 size=128
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=1 messageSize=256 iterations=35
PingPong: otherRank 1, total time 63.223 us, loop 35, bufLen 256, latency 0.903
us. bandwidth 0.283441 GB/s
Simulation is complete, simulated time: 71.5131 us
```

```
=== adjacent on mesh 16x16 (PPN=2) ===
==================================================
 Pattern           : adjacent
 Topology          : Mesh 16x16  (256 routers, 512 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 1
 Representative hops: 0  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/adjacent/results/adjacent
_mesh_16x16_ppn2_nodes512_stats.csv
==================================================
180000:0:1:EmberEngine:PingPong:generate:65: rank=1 size=512
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=512
rank2=1 messageSize=256 iterations=35
PingPong: otherRank 1, total time 63.223 us, loop 35, bufLen 256, latency 0.903
us. bandwidth 0.283441 GB/s
Simulation is complete, simulated time: 76.0899 us
=== adjacent on mesh 32x32 (PPN=2) ===
==================================================
 Pattern           : adjacent
 Topology          : Mesh 32x32  (1024 routers, 2048 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 1
 Representative hops: 0  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/adjacent/results/adjacent
_mesh_32x32_ppn2_nodes2048_stats.csv
==================================================
180000:0:1:EmberEngine:PingPong:generate:65: rank=1 size=2048
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=2048
rank2=1 messageSize=256 iterations=35
PingPong: otherRank 1, total time 63.313 us, loop 35, bufLen 256, latency 0.904
us. bandwidth 0.283038 GB/s
Simulation is complete, simulated time: 83.7539 us
=== halfshift on mesh 2x2 (PPN=2) ===
==================================================
 Pattern           : halfshift
 Topology          : Mesh 2x2  (4 routers, 8 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 4
 Representative hops: 1  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/halfshift/results/halfshi
ft_mesh_2x2_ppn2_nodes8_stats.csv
==================================================
180000:0:4:EmberEngine:PingPong:generate:65: rank=4 size=8
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=4 messageSize=256 iterations=35
PingPong: otherRank 4, total time 69.271 us, loop 35, bufLen 256, latency 0.990
us. bandwidth 0.258694 GB/s
Simulation is complete, simulated time: 73.1867 us
=== halfshift on mesh 4x4 (PPN=2) ===
==================================================
 Pattern           : halfshift
 Topology          : Mesh 4x4  (16 routers, 32 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 16
 Representative hops: 2  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
```

```
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/halfshift/results/halfshi
ft_mesh_4x4_ppn2_nodes32_stats.csv
==================================================
180000:0:16:EmberEngine:PingPong:generate:65: rank=16 size=32
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=16 messageSize=256 iterations=35
PingPong: otherRank 16, total time 75.319 us, loop 35, bufLen 256, latency 1.076
 us. bandwidth 0.237921 GB/s
Simulation is complete, simulated time: 83.4715 us
=== halfshift on mesh 8x8 (PPN=2) ===
==================================================
 Pattern           : halfshift
 Topology          : Mesh 8x8  (64 routers, 128 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 64
 Representative hops: 4  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/halfshift/results/halfshi
ft_mesh_8x8_ppn2_nodes128_stats.csv
==================================================
180000:0:64:EmberEngine:PingPong:generate:65: rank=64 size=128
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=64 messageSize=256 iterations=35
PingPong: otherRank 64, total time 87.415 us, loop 35, bufLen 256, latency 1.249
 us. bandwidth 0.204999 GB/s
Simulation is complete, simulated time: 100.603 us
=== halfshift on mesh 16x16 (PPN=2) ===
==================================================
 Pattern           : halfshift
 Topology          : Mesh 16x16  (256 routers, 512 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 256
 Representative hops: 8  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/halfshift/results/halfshi
ft_mesh_16x16_ppn2_nodes512_stats.csv
==================================================
180000:0:256:EmberEngine:PingPong:generate:65: rank=256 size=512
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=512
rank2=256 messageSize=256 iterations=35
PingPong: otherRank 256, total time 111.607 us, loop 35, bufLen 256, latency 1.5
94 us. bandwidth 0.160563 GB/s
Simulation is complete, simulated time: 132.444 us
=== halfshift on mesh 32x32 (PPN=2) ===
==================================================
 Pattern           : halfshift
 Topology          : Mesh 32x32  (1024 routers, 2048 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 1024
 Representative hops: 16  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/halfshift/results/halfshi
ft_mesh_32x32_ppn2_nodes2048_stats.csv
==================================================
180000:0:1024:EmberEngine:PingPong:generate:65: rank=1024 size=2048
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=2048
rank2=1024 messageSize=256 iterations=35
```

```
PingPong: otherRank 1024, total time 159.991 us, loop 35, bufLen 256, latency 2.
286 us. bandwidth 0.112006 GB/s
Simulation is complete, simulated time: 192.857 us
=== mirrored on mesh 2x2 (PPN=2) ===
==================================================
 Pattern           : mirrored
 Topology          : Mesh 2x2  (4 routers, 8 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 7
 Representative hops: 2  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/mirrored/results/mirrored
_mesh_2x2_ppn2_nodes8_stats.csv
==================================================
180000:0:7:EmberEngine:PingPong:generate:65: rank=7 size=8
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=7 messageSize=256 iterations=35
PingPong: otherRank 7, total time 75.319 us, loop 35, bufLen 256, latency 1.076
us. bandwidth 0.237921 GB/s
Simulation is complete, simulated time: 80.0779 us
=== mirrored on mesh 4x4 (PPN=2) ===
==================================================
 Pattern           : mirrored
 Topology          : Mesh 4x4  (16 routers, 32 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 31
 Representative hops: 6  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/mirrored/results/mirrored
_mesh_4x4_ppn2_nodes32_stats.csv
==================================================
180000:0:31:EmberEngine:PingPong:generate:65: rank=31 size=32
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=31 messageSize=256 iterations=35
PingPong: otherRank 31, total time 99.511 us, loop 35, bufLen 256, latency 1.422
 us. bandwidth 0.180081 GB/s
Simulation is complete, simulated time: 107.988 us
=== mirrored on mesh 8x8 (PPN=2) ===
==================================================
 Pattern           : mirrored
 Topology          : Mesh 8x8  (64 routers, 128 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 127
 Representative hops: 14  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/mirrored/results/mirrored
_mesh_8x8_ppn2_nodes128_stats.csv
==================================================
180000:0:127:EmberEngine:PingPong:generate:65: rank=127 size=128
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=127 messageSize=256 iterations=35
PingPong: otherRank 127, total time 147.895 us, loop 35, bufLen 256, latency 2.1
13 us. bandwidth 0.121167 GB/s
Simulation is complete, simulated time: 160.716 us
=== mirrored on mesh 16x16 (PPN=2) ===
==================================================
 Pattern           : mirrored
 Topology          : Mesh 16x16  (256 routers, 512 nodes, PPN=2)
```

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 Exchange          : rank 0  <->  rank 511
 Representative hops: 30  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/mirrored/results/mirrored
_mesh_16x16_ppn2_nodes512_stats.csv
==================================================
180000:0:511:EmberEngine:PingPong:generate:65: rank=511 size=512
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=512
rank2=511 messageSize=256 iterations=35
PingPong: otherRank 511, total time 244.663 us, loop 35, bufLen 256, latency 3.4
95 us. bandwidth 0.073244 GB/s
Simulation is complete, simulated time: 263.752 us
=== mirrored on mesh 32x32 (PPN=2) ===
==================================================
 Pattern           : mirrored
 Topology          : Mesh 32x32  (1024 routers, 2048 nodes, PPN=2)
 Exchange          : rank 0  <->  rank 2047
 Representative hops: 62  (Manhattan dist between routers)
 PPN (ranks/router): 2   <-- should be 2
 Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/C
ommunication _Pattern/Communication_Pattern_PPN1/PPN_2/mirrored/results/mirrored
_mesh_32x32_ppn2_nodes2048_stats.csv
==================================================
180000:0:2047:EmberEngine:PingPong:generate:65: rank=2047 size=2048
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=2048
rank2=2047 messageSize=256 iterations=35
PingPong: otherRank 2047, total time 438.199 us, loop 35, bufLen 256, latency 6.
260 us. bandwidth 0.040895 GB/s
Simulation is complete, simulated time: 466.552 us
All PPN=2 pattern runs finished.
-rw-r--r--  1 stephenrajva  staff   916408 28 Jun 07:22 adjacent/results/adjacen
t_mesh_16x16_ppn2_nodes512_stats.csv
-rw-r--r--  1 stephenrajva  staff    11392 28 Jun 07:21 adjacent/results/adjacen
t_mesh_2x2_ppn2_nodes8_stats.csv
-rw-r--r--  1 stephenrajva  staff  3756241 28 Jun 07:22 adjacent/results/adjacen
t_mesh_32x32_ppn2_nodes2048_stats.csv
-rw-r--r--  1 stephenrajva  staff    51749 28 Jun 07:22 adjacent/results/adjacen
t_mesh_4x4_ppn2_nodes32_stats.csv
-rw-r--r--  1 stephenrajva  staff   219997 28 Jun 07:22 adjacent/results/adjacen
t_mesh_8x8_ppn2_nodes128_stats.csv
-rw-r--r--  1 stephenrajva  staff    11445 28 Jun 07:21 adjacent/results/mirrore
d_mesh_2x2_ppn2_nodes8_stats.csv
-rw-r--r--  1 stephenrajva  staff   933480 28 Jun 07:22 halfshift/results/halfsh
ift_mesh_16x16_ppn2_nodes512_stats.csv
-rw-r--r--  1 stephenrajva  staff    11412 28 Jun 07:22 halfshift/results/halfsh
ift_mesh_2x2_ppn2_nodes8_stats.csv
-rw-r--r--  1 stephenrajva  staff  3825336 28 Jun 07:22 halfshift/results/halfsh
ift_mesh_32x32_ppn2_nodes2048_stats.csv
-rw-r--r--  1 stephenrajva  staff    51794 28 Jun 07:22 halfshift/results/halfsh
ift_mesh_4x4_ppn2_nodes32_stats.csv
-rw-r--r--  1 stephenrajva  staff   223173 28 Jun 07:22 halfshift/results/halfsh
ift_mesh_8x8_ppn2_nodes128_stats.csv
-rw-r--r--  1 stephenrajva  staff   933953 28 Jun 07:22 mirrored/results/mirrore
d_mesh_16x16_ppn2_nodes512_stats.csv
-rw-r--r--  1 stephenrajva  staff    11445 28 Jun 07:22 mirrored/results/mirrore
d_mesh_2x2_ppn2_nodes8_stats.csv
-rw-r--r--  1 stephenrajva  staff  3834072 28 Jun 07:22 mirrored/results/mirrore
```

```
d_mesh_32x32_ppn2_nodes2048_stats.csv
-rw-r--r--  1 stephenrajva  staff    52806 28 Jun 07:22 mirrored/results/mirrore
d_mesh_4x4_ppn2_nodes32_stats.csv
-rw-r--r--  1 stephenrajva  staff   224398 28 Jun 07:22 mirrored/results/mirrore
d_mesh_8x8_ppn2_nodes128_stats.csv
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP
C-Upload 2/SST_Stimulator/Communication _Pattern/PPN_2"
rm -f adjacent/results/mirrored_mesh_2x2_ppn2_nodes8_stats.csv
echo "adjacent:";  ls adjacent/results/*.csv  | wc -l
echo "halfshift:"; ls halfshift/results/*.csv | wc -l
echo "mirrored:";  ls mirrored/results/*.csv  | wc -l
adjacent:
       5
halfshift:
       5
mirrored:
       5
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP
C-Upload 2/SST_Stimulator/Communication _Pattern/PPN_2"
cp "../Communication_Pattern_PPN1/analyze_patterns.py" .
cat analyze_patterns.py
#!/usr/bin/env python3
# ==========================================================================
#  analyze_patterns.py  -  Compare adjacent / halfshift / mirrored on a Mesh.
#  Reads each pattern run's NIC packet_latency from its CSV and pairs it with
#  the representative Manhattan hop count, then overlays all three patterns on
#  two charts: latency-vs-size and hops-vs-size.
#
#  Run from the Communication_Pattern_PPN1 folder (with adjacent/ halfshift/
#  mirrored/ subfolders each containing results/).
# ==========================================================================
import csv, os
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

PATTERNS = ["adjacent", "halfshift", "mirrored"]
# (label, shape, nodes)
SIZES = [("2x2", 4), ("4x4", 16), ("8x8", 64), ("16x16", 256), ("32x32", 1024)]

def partner_of(pattern, N):
    if pattern == "adjacent":  return 1
    if pattern == "halfshift": return N // 2
    if pattern == "mirrored":  return N - 1

def coord(nid, dims):
    c, rem = [], nid
    for d in dims:
        c.append(rem % d); rem //= d
    return c

def hops(pattern, shape, N):
    dims = [int(x) for x in shape.split("x")]
    p = partner_of(pattern, N)
    a, b = coord(0, dims), coord(p, dims)
    return sum(abs(x - y) for x, y in zip(a, b))

def avg_latency_ns(csv_path):
```

```
    """Average packet_latency (ns) over NIC rows that recorded it."""
    if not os.path.exists(csv_path):
        return None
    tot_sum, tot_cnt = 0.0, 0
    with open(csv_path) as f:
        for row in csv.reader(f):
            row = [c.strip() for c in row]
            if len(row) < 9:
                continue
            if row[1] == "packet_latency":
                try:
                    s = float(row[6]); c = float(row[8])
                except ValueError:
                    continue
                tot_sum += s; tot_cnt += c
    if tot_cnt == 0:
        return None
    return tot_sum / tot_cnt

# gather
data = {p: {"nodes": [], "lat": [], "hop": []} for p in PATTERNS}
print(f"{'Pattern':10} {'Size':7} {'Nodes':6} {'Hops':5} {'Latency(ns)':12}")
print("-" * 50)
for p in PATTERNS:
    for shape, N in SIZES:
        csv_path = os.path.join(p, "results", f"{p}_mesh_{shape}_ppn1_nodes{N}_s
tats.csv")
        lat = avg_latency_ns(csv_path)
        h = hops(p, shape, N)
        if lat is None:
            print(f"{p:10} {shape:7} {N:6} {h:5}  MISSING ({csv_path})")
            continue
        data[p]["nodes"].append(N)
        data[p]["lat"].append(lat)
        data[p]["hop"].append(h)
        print(f"{p:10} {shape:7} {N:6} {h:5} {lat:12.2f}")

# latency vs size
plt.figure(figsize=(8, 5))
for p in PATTERNS:
    plt.plot(data[p]["nodes"], data[p]["lat"], marker="o", label=p)
plt.xscale("log", base=2)
plt.xlabel("Nodes (PPN=1)"); plt.ylabel("Avg packet latency (ns)")
plt.title("Communication patterns on Mesh - Latency vs Size")
plt.legend(); plt.grid(True, which="both", ls=":")
plt.savefig("pattern_latency_vs_size.png", dpi=130, bbox_inches="tight")
print("Saved: pattern_latency_vs_size.png")

# hops vs size
plt.figure(figsize=(8, 5))
for p in PATTERNS:
    plt.plot(data[p]["nodes"], data[p]["hop"], marker="s", label=p)
plt.xscale("log", base=2)
plt.xlabel("Nodes (PPN=1)"); plt.ylabel("Representative hops (Manhattan)")
plt.title("Communication patterns on Mesh - Hops vs Size")
plt.legend(); plt.grid(True, which="both", ls=":")
plt.savefig("pattern_hops_vs_size.png", dpi=130, bbox_inches="tight")
print("Saved: pattern_hops_vs_size.png")
```

```
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % >....
by 2)
s = s.replace(
    '''    p = partner_of(pattern, N)
    a, b = coord(0, dims), coord(p, dims)
    return sum(abs(x - y) for x, y in zip(a, b))''',
    '''    p = partner_of(pattern, N)
    a, b = coord(0 // 2, dims), coord(p // 2, dims)   # PPN=2: node id // 2 = ro
uter id
    return sum(abs(x - y) for x, y in zip(a, b))'''
)

# 4. labels/titles -> PPN=2
s = s.replace('Nodes (PPN=1)', 'Nodes (PPN=2)')
s = s.replace('Mesh - Latency vs Size', 'Mesh - Latency vs Size (PPN=2)')
s = s.replace('Mesh - Hops vs Size', 'Mesh - Hops vs Size (PPN=2)')
s = s.replace('pattern_latency_vs_size.png', 'pattern_latency_vs_size_ppn2.png')

s = s.replace('pattern_hops_vs_size.png', 'pattern_hops_vs_size_ppn2.png')

open(f, "w").write(s)
print("Updated analyze_patterns.py for PPN=2.")
EOF
python3 analyze_patterns.py
Updated analyze_patterns.py for PPN=2.
Pattern    Size    Nodes  Hops  Latency(ns)
--------------------------------------------------
adjacent   2x2          8     0       126.18
adjacent   4x4         32     0       198.44
adjacent   8x8        128     0       396.22
adjacent   16x16      512     0       777.85
adjacent   32x32     2048     0      1493.60
halfshift  2x2          8     1       198.23
halfshift  4x4         32     2       290.30
halfshift  8x8        128     4       470.87
halfshift  16x16      512     8       822.18
halfshift  32x32     2048    16      1516.88
mirrored   2x2          8     2       270.46
mirrored   4x4         32     6       473.27
mirrored   8x8        128    14       657.57
mirrored   16x16      512    30       944.02
mirrored   32x32     2048    62      1583.69
Saved: pattern_latency_vs_size_ppn2.png
Saved: pattern_hops_vs_size_ppn2.png
stephenrajva@Stephens-MacBook-Air-4 PPN_2 %
```